

Multi-Agent Template Filling using LLMs

Design and Comparative Evaluation with Speech Input Case Study

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DATE

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The Problem: From Speech to Structured Reports



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 UNSTRUCTURED SPEECH INPUT

"We had a minor safety incident near the blast furnace around 2 p.m.—someone tripped, but no injuries. Conveyor belt on Line B stopped twice. Production target met—482 tons."

 STRUCTURED TEMPLATE OUTPUT

Safety Incident	Minor, near blast furnace
Time	~2 PM
Equipment Issue	Line B conveyor belt
Production	482 tons

Operator
Speaks




Audio
Transcription




AI
Extraction




Structured
Template

Why Current Systems Fall Short

Fragile

Single-model systems fail on noisy, incomplete, or reordered speech input.

Opaque

No error attribution — impossible to trace if failure is in transcription, understanding, or mapping.

Rigid

Cannot adapt to new domains or schemas without costly retraining.

Research Objectives



Objective 1: Design

Design a modular **multi-agent architecture** that separates template filling into five specialized subtasks:

✓ Transcription (STT)

✓ Retrieval (RAG)

✓ Extraction (IE)

✓ Formatting (CF)

✓ Verification (VER)



Objective 2: Evaluate

Empirical evaluation using the **MUC-4 benchmark dataset** to measure performance across:

✓ Extraction Accuracy

✓ Latency

✓ Consistency

✓ Cost

RESEARCH SCOPE

⊘ No Model Fine-tuning

⚙️ System Orchestration & Prompt Engineering

📄 MUC-4 (N=100)

Requirements for a Robust Template Filling System

Consistency

Normalize inputs into uniform date, terminology, and style formats.

Extraction

Extract relevant details from noisy input; leave irrelevant fields empty.

Transparency

Show confidence scores for every extracted value.

User Correction

Allow manual editing of any field before final submission.

Adaptation

Improve over time using historical templates via RAG.

Usability

Achieve near real-time response times for production use.

Existing Approaches Leave Critical Gaps

Template Filling & Extraction	Prompt Engineering	Multi-Agent Systems
Template Filling with Generative Transformers [1] <i>Du et al. · 2021</i>	In-context Learning with Few-Shot Prompting [5] <i>Brown et al. · 2020</i>	HuggingGPT: Solving AI Tasks with ChatGPT [8] <i>Shen et al. · 2023</i>
Speech-based Slot Filling using Large Language Models [2] <i>Sun et al. · 2023</i>	Speech-based Slot Filling using Large Language Models [2] <i>Sun et al. · 2023</i>	TaskMatrix.AI: Multi-Agent Task Solving [9] <i>Liang et al. · 2023</i>
Web App Test Generation via Multi-Agent LLMs [3] <i>Le et al. · 2025</i>	SGP-TOD: Building Task Bots via Schema-Guided Prompting [6] <i>Zhang et al. · 2023</i>	Cooperative Multi-Agent NER Framework [10] <i>Wang et al. · 2025</i>
LangExtract: Schema-based IE with LLMs [4] <i>Google Research · 2024</i>	Automated Prompt Optimization Framework [7] <i>Peng et al. · 2023</i>	AutoAgents: Consensus-Based Multi-Agent Framework [11] <i>Wu et al. · 2023</i>

Existing Approaches Leave Critical Gaps

Solution / Approach	R1 CONSISTENCY	R2 EXTRACTION	R3 TRANSPARENCY	R4 USER CORRECTION	R5 ADAPTATION	R6 USABILITY
Template Filling with Generative Transformers [1]						
Speech-based Slot Filling using Large Language Models [2]						
Auto Agents: An Open-Source Framework for Building Multi-Agent Systems with Large Language Models. [11]						
Google Research: LangExtract: Schema-based Information Extraction with Large Language Models. [4]						

 **High**  **Medium**  **Low**

 **Not-Mentioned**

THE CRITICAL GAP

No existing system provides transparency + user correction + learning + usability together.

Invox: A Five-Agent Modular Pipeline



"Each agent has a single responsibility, clear inputs/outputs, and can be independently upgraded."

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Four Ways to Orchestrate the Pipeline

S1 Single-Pass Full Input

Processes all template fields in a single LLM request. Ideal for high-throughput, low-cost scenarios.

INPUT → LLM (ALL FIELDS) → OUTPUT

⚡ FASTEST 💰 CHEAPEST

S2 Iterative Per-Field

Sequential or parallel calls for each individual field. Isolates errors and prevents prompt-length degradation.

INPUT →
 LLM (FIELD 1)
 LLM (FIELD 2)
 LLM (FIELD N) → OUTPUT

🛡️ ERROR ISOLATION 🔄 PARALLELIZABLE

S3 Multi-LLM Consensus (Full)

Uses heterogeneous models (e.g., GPT-4 + Gemini) for full processing. A third LLM acts as a judge.

INPUT →
 LLM A (All FIELDS)
 LLM B (All FIELDS) → JUDGE → OUTPUT

🛡️ ROBUST 🗳️ CONSENSUS

S4 Multi-LLM Consensus (Per-Field)

Highest level of granularity. Consensus mechanism applied to every single slot in the template.

INPUT →
 LLM A (FIELD 1)
 LLM B (FIELD 1)
 ...
 LLM A (FIELD N)
 LLM B (FIELD N) → JUDGE → OUTPUT

💎 MAX ACCURACY 🏠 HIGH COST

Four Ways to Orchestrate the Pipeline

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Processes all template fields in a single LLM request. Ideal for high-throughput, low-cost scenarios.

INPUT



LLM (ALL FIELDS)



OUTPUT

 **FASTEST**  **CHEAPEST**

Four Ways to Orchestrate the Pipeline

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 **ERROR ISOLATION**  **PARALLELIZABLE**

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📈 **MAX ACCURACY** 💰 **HIGH COST**

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 LLM B (All FIELDS) → JUDGE → OUTPUT

🛡️ ROBUST 🗳️ CONSENSUS

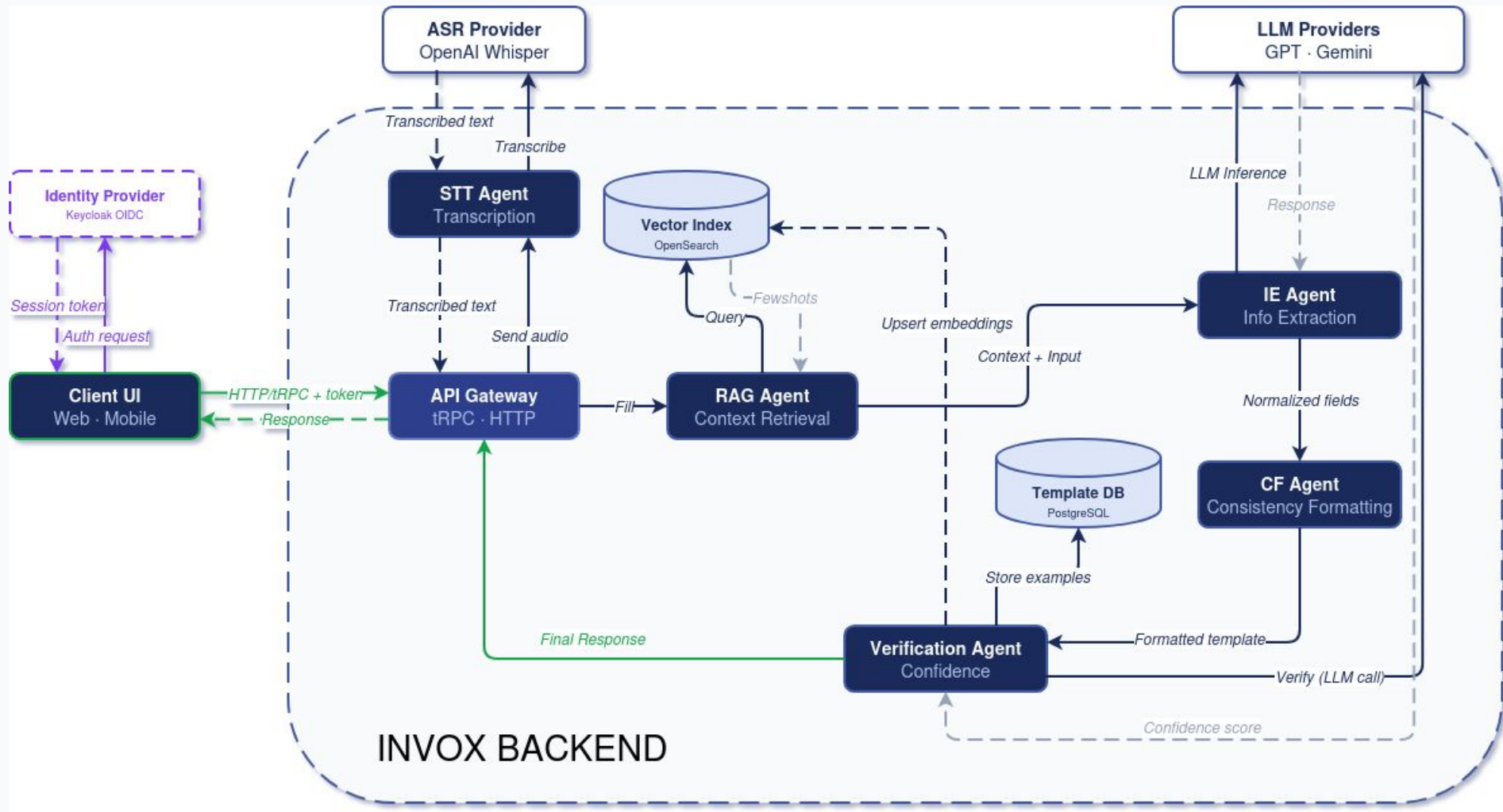
S4 Multi-LLM Consensus (Per-Field)

Highest level of granularity. Consensus mechanism applied to every single slot in the template.

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 LLM A (FIELD 1)
 LLM B (FIELD 1)
 ...
 LLM A (FIELD N)
 LLM B (FIELD N) → JUDGE → OUTPUT

💎 MAX ACCURACY 🏠 HIGH COST

System Architecture & Workflow



Implementation Details

SC

Assistant

Hi! Click **Start** to record, then **Stop & Process**. You'll see *Transcribing...* and *Filling...* while I work—then I'll fill the form. Describe the incident in your own words.

You

Hello, hello, this is Muzaffar Jadliwala, today is 5th of January, 2026 and it is a morning shift. I am the shift leader and from now on, like, I would like to tell that the shift was fine except the production line 5 was not working. It should be fixed by next shift and like order number 543 is under process. Order number 542 is already done and dispatched. So this is all.

Assistant

I updated the date to January 5th, 2026, and added you as the shift leader, Muzaffar Jadliwala. The shift was set to the morning shift (FRÜHSCHICHT). I also included the status of the orders, noting that order 543 is in progress while order 542 is completed and dispatched. Additionally, I recorded that production line 5 is currently down and scheduled for repair by the next shift, with the technical team responsible. If there are any other details you'd like to add or adjust, just let me know!

Language:

EnglishGerman

Start Recording

Aiconver Admin

Shift Handover Report

Datum ⓘ

01 / 05 / 2026

Schichtleiter ⓘ

Muzaffar Jadliwala

Schicht ⓘ

FRÜHSCHICHT

Status der laufenden Aufträge/Projekte ⓘ

• 543 – in Bearbeitung; • 542 – abgeschlossen und versendet.

Wichtige Ereignisse während der Schicht

Offene Aufgaben ⓘ

• Produktionslinie 5 reparieren – Priorität: Hoch – Verantwortlich: Technik (T. Müller) – Fällig: nächste Schicht – Abhängigkeit: Teile verfügbar.

Maschinen-/Systemstatus ⓘ

• Produktionslinie 5: Ausfall; Maßnahmen: Reparatur geplant.

Besondere Vorkommnisse

Shift Handover Report Download

JSON

Excel

PDF



Web App

How We Evaluated

Dataset: MUC-4 Terrorism Event Extraction Benchmark [12]



BENCHMARK DATA

Total Articles	1,300
Test Set (N)	100 Documents



Semantic Embedding-Based Metrics

Traditional exact match penalizes valid paraphrases (e.g., "guerrillas" vs "rebel forces"). We employ **all-MiniLM-L6-v2** sentence embeddings + **cosine similarity** to capture true intent.



9 EXTRACTED FIELDS

incidentType, incidentDate

incidentLocation, incidentStage

target, victim, weapon

perpetratorIndividual

perpetratorOrganization

OBS / NES

Overall Baseline vs. Non-Empty Score (Accuracy)

SF1 (Soft F1)

Harmonic mean of soft coverage and specificity

FDA

Fill-Decision Accuracy (Abstention Quality)

HR / MR

Hallucination Rate vs. Missing Rate

EAI

OBS minus NES

FPR / SC

Format Preservation Rate and Style Consistency

Three Evaluation Scenarios

1

BASELINE SCENARIO

Text Only (No RAG)

MUC-4 text is passed directly to the IE agent with no retrieval augmentation. This establishes the performance floor — what the system can do with prompting alone.



MUC-4 Text Input



No RAG / No Few-Shot



No Speech Simulation

Purpose: Establishes baseline performance

2

MAIN EVALUATION

★ PRIMARY FOCUS

Text + RAG (Few-Shot .1)

MUC-4 text with RAG-retrieved few-shot examples injected into the prompt. This is the primary evaluation scenario — all results in the following slides refer to this variant.



MUC-4 Text Input



RAG Enabled — Few-Shot Examples



No Speech Simulation

 All following result slides use this scenario

3

ROBUSTNESS TEST

Simulated Speech + RAG

An LLM rewrites MUC-4 text as natural spoken language — informal phrasing, topic reordering, hesitations. Tests system robustness to real-world noisy speech input.



LLM-Simulated Speech Text



RAG Enabled — Few-Shot Examples



Noise, Reordering & Informality

Purpose: Tests robustness requirement (R2, R6)

Extraction



Consistency, Transparency, and User Correction

FPR — Format Preservation Rate

↑ Higher is better

Does the output follow the correct schema format?



SC — Style Consistency

↑ Higher is better

Character 3-gram cosine similarity — style uniformity



R3 — Transparency

Every extracted field shows a confidence score and source evidence — enabling the operator to verify any value before submission.

✓ Satisfied — confidence scores visible in UI



R4 — User Correction

All template fields are fully editable before final submission. The AI fills — the human decides. No field is locked or auto-submitted.

✓ Satisfied — all fields editable before submission

■ S1 — Single-Pass ■ S2 — Iterative ■ S3 — Consensus Full ■ S4 — Consensus Per-Field

All values: few-shot ,1 variants · MUC-4 N=100

Adaptation and Usability

R5 — Adaptation: RAG Impact on OBS

OBS improvement from no few-shot (.0) → with RAG (.1)

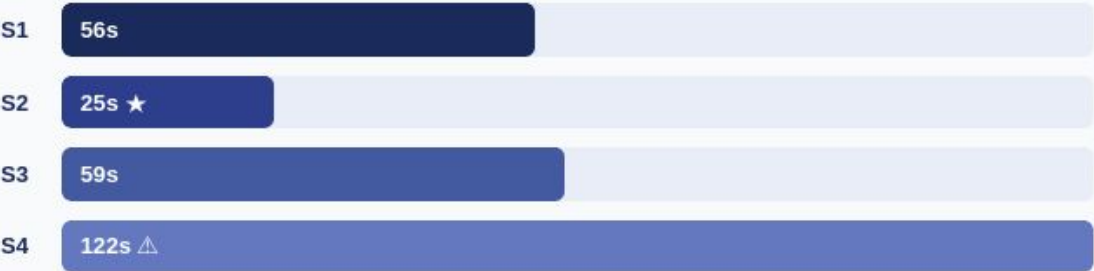
↑ Improvement



R6 — Usability: Median Latency (p50)

Median response time per document in seconds

↓ Lower is better



R5 — Adaptation ✓

RAG consistently improves OBS across all four strategies. Largest gain in S1 (+7.2%). Few-shot examples from historical templates directly boost extraction accuracy.

✓ RAG improves accuracy across all strategies

R6 — Cost per Document



✓ vs ~\$10 manual cost — 99% cost reduction

What We Learned



No Single Dominant Strategy

Deployment choice depends on constraints:

S1: Best All-Rounder | **S2:** Fastest (25s)

S3: Most Robust | **S4:** Highest Recall



The "RAG Boost" is Consistent

Few-shot retrieved examples significantly enhanced all metrics across every strategy tested:

~ **+6%**

ACCURACY GAIN

~ **-6.0%**

HALLUCINATION REDUCTION



SPEECH ROBUSTNESS

Speech-style transcripts caused only **~3% degradation**. Modular agents absorbed noise effectively.



METRIC VALIDITY

Soft-F1 (Embeddings) captured semantic meaning where exact-match failed on valid synonyms.



ECONOMIC IMPACT

Around **€ 0.72 per document** (S1). Massive ROI compared to ~~\$10~~ estimated manual labor cost.

Limitations & Next Steps

Current Limitations

 *Domain Specificity*

 *Evaluation Depth*

 *Input Modeling*

 *Ablation Gap*

Future Work Directions

 *Real-World Validation*

 *Adaptive Routing*

 *Multimodal Expansion*

 *Model Distillation*

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Thank You



Questions & Discussion

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DEFENSE DATE

March 18, 2026

Scenario 1: Text Only (No RAG)

APPENDIX

Strategy	R1 — Consistency		R2 — Extraction Accuracy						R3	R4	R5	R6 — Usability	
	FPR ↑	SC ↑	OBS ↑	NES ↑	SF1 ↑	FDA ↑	HR ↓	MR ↓	Transparency	User Correction	Adaptation	Latency (p50)	Cost/doc
S1 — Single-Pass	0.910	0.692	0.601	0.469	0.604	0.713	0.233	0.329	✓	✓	✓	25.0s	0.72¢
S2 — Iterative	0.970	0.672	0.590	0.560	0.608	0.709	0.373	0.226	✓	✓	✓	20.0s	2.18¢
S3 — Consensus Full	0.910	0.691	0.602	0.481	0.601	0.713	0.246	0.319	✓	✓	✓	31.5s	1.40¢
S4 — Consensus Per-Field	0.870	0.660	0.549	0.628	0.589	0.693	0.551	0.112	✓	✓	✓	125.1s	4.32¢

Scenario 2:Text + RAG

APPENDIX

Strategy	R1 — Consistency		R2 — Extraction Accuracy						R3	R4	R5	R6 — Usability	
	FPR ↑	SC ↑	OBS ↑	NES ↑	SF1 ↑	FDA ↑	HR ↓	MR ↓	Transparency	User Correction	Adaptation	Latency (p50)	Cost/doc
S1 — Single-Pass	0.960	0.724	0.644	0.504	0.642	0.746	0.180	0.313	✓	✓	✓	55.9s	0.72¢
S2 — Iterative	0.950	0.683	0.619	0.555	0.639	0.718	0.301	0.267	✓	✓	✓	25.4s	2.18¢
S3 — Consensus Full	0.920	0.724	0.641	0.521	0.638	0.743	0.208	0.295	✓	✓	✓	59.4s	1.40¢
S4 — Consensus Per-Field	0.940	0.688	0.579	0.624	0.613	0.709	0.476	0.144	✓	✓	✓	121.8s	4.32¢

Scenario 3: Simulated Speech + RAG

APPENDIX

Strategy	R1 — Consistency		R2 — Extraction Accuracy						R3	R4	R5	R6 — Usability	
	FPR ↑	SC ↑	OBS ↑	NES ↑	SF1 ↑	FDA ↑	HR ↓	MR ↓	Transparency	User Correction	Adaptation	Latency (p50)	Cost/doc
S1 — Single-Pass	0.930	0.700	0.626	0.508	0.631	0.727	0.226	0.311	✓	✓	✓	49.4s	0.72¢
S2 — Iterative	0.940	0.666	0.595	0.544	0.631	0.700	0.341	0.267	✓	✓	✓	24.1s	2.18¢
S3 — Consensus Full	0.910	0.702	0.624	0.524	0.630	0.728	0.251	0.289	✓	✓	✓	53.5s	1.40¢
S4 — Consensus Per-Field	0.940	0.682	0.572	0.612	0.592	0.704	0.479	0.150	✓	✓	✓	127.2s	4.32¢

Trade-offs at a Glance

APPENDIX

STRATEGY	LLM CALLS	ERROR ISOLATION	ROBUSTNESS	LATENCY	COST
S1: Single-Pass	1	LOW	LOW	LOW	LOW
S2: Iterative	n	HIGH	MEDIUM	LOW	MEDIUM
S3: Consensus Full	m	LOW	HIGH	MEDIUM	HIGH
S4: Consensus Per-Field	$n \times m$	HIGH	HIGH	HIGH	HIGH

Evaluation Matrix

APPENDIX

OBS Overall Baseline Score

Accuracy across ALL 9 fields including empty ones. If gold is empty AND system leaves it empty — that counts as correct too.

Example:

Gold: victim=(empty). System: victim=(empty). → OBS gets 1.0 for this field.
Easy win for cautious systems.

NES Non-Empty Score

Accuracy ONLY on fields where the gold has a real value. Empty fields are completely ignored. The harder, more honest test.

Example:

Gold: weapon='bomb'. System: weapon='rifle'. → NES gets ~0.3. System tried but got it wrong. No hiding behind empty fields.

EAI Empty Advantage Index

Simply OBS minus NES. A high EAI means the system's good OBS score comes mostly from correctly leaving things empty — not real extraction.

Example:

LangExtract: OBS=0.684, NES=0.406 → EAI=0.278. High! Most of its score is from empty fields, not real extracted values.

SF1 Soft F1

Harmonic mean of Coverage (did you find all values?) and Specificity (was everything found real?). Catches both missing AND hallucinated values.

Example:

Gold: weapon=['bomb','rifle']. Pred: ['bomb','rifle','grenade'] → Coverage=1.0, Specificity=0.67 → SF1=0.80. Penalises fake 'grenade'.

Evaluation Matrix

APPENDIX

FDA Fill Decision Accuracy

Did the system make the right YES/NO choice — fill or leave empty? Does NOT care about the value quality, only the decision itself.

Example:

Gold: location='Bogotá'. System: location='Colombia'. FDA=1.0 — correct decision to fill, even though the value is slightly wrong.

HR Hallucination Rate ↓ lower is better

How often does the system fill a field that should be EMPTY? The system is inventing values not supported by the text.

Example:

S4: HR=0.476 — nearly half of all empty fields get a fake value filled in. That is the big red bar on slide 21.

MR Missing Rate ↓ lower is better

How often does the system leave a field EMPTY when there IS a real value in the text? System is being too cautious.

Example:

LangExtract: MR=0.369 — misses over a third of all real values. Very safe system but not useful for operators.

FPR Format Preservation Rate

Is the output in the right format? Correct date format (YYYY-MM-DD), valid enum values, proper JSON structure. Your CF agent handles this.

Example:

System outputs 'May 15 1990' for incidentDate → FPR fails. System outputs '1990-05-15' → FPR passes. LangExtract scores a perfect 1.0.

SC Style Consistency

Are similar values written in the same style across all 100 documents? Not just correct — but uniform. Measured with character n-gram similarity.

Example:

Doc 1: 'Bogotá, Colombia'. Doc 2: 'colombia'. Doc 3: 'COLOMBIA'. All technically valid — but SC would be low due to inconsistent style.